Pushing on Text Readability Assessment: A Transformer Meets Handcrafted Linguistic Features

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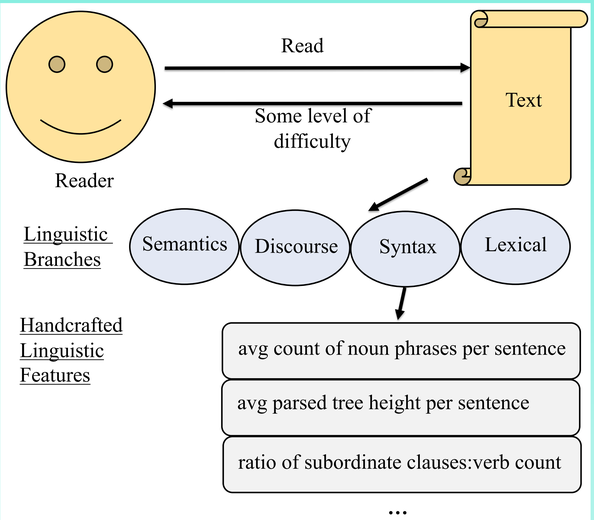
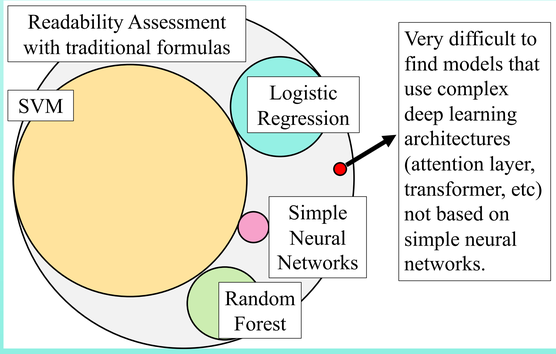
1 - Abstract

We report two essential improvements in readability assessment: 1. three novel features in advanced semantics and 2. the timely evidence that traditional ML models (e.g. Random Forest, using handcrafted features) can combine with transformers (e.g. RoBERTa) to augment model performance. First, we explore suitable transformers and traditional ML models. Then, we extract 255 handcrafted linguistic features using self-developed extraction software. Finally, we assemble those to create several hybrid models, achieving state-of-the-art (SOTA) accuracy on popular datasets in readability assessment. The use of handcrafted features help model performance on smaller datasets. Notably, our RoBERTA-RF-T1 hybrid achieves the near-perfect classification accuracy of 99%, a 20.3% increase from the previous SOTA.

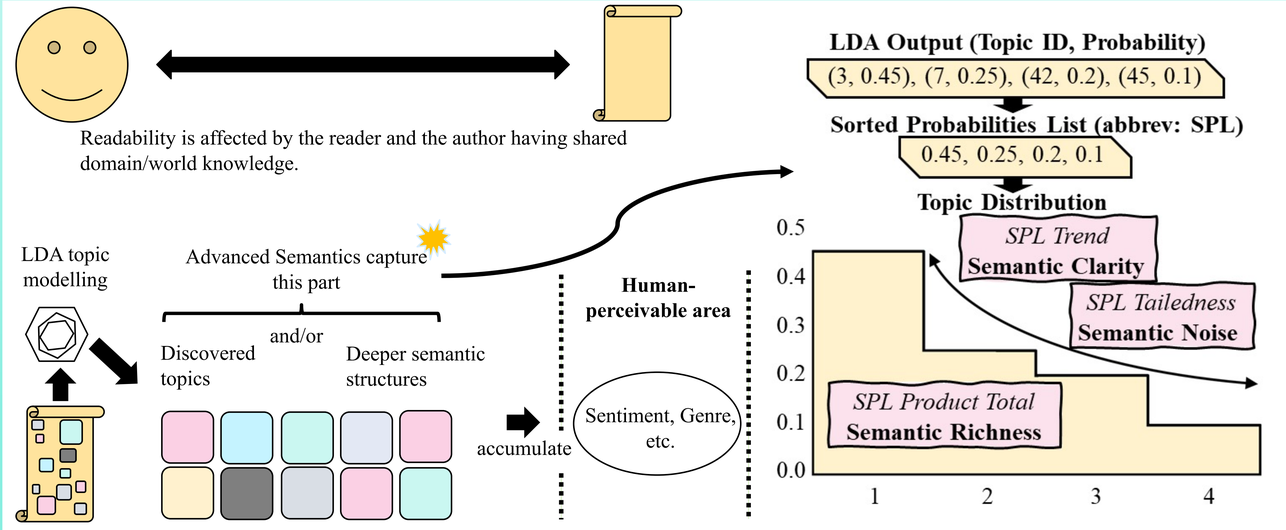
2 - What is Readability Assess.



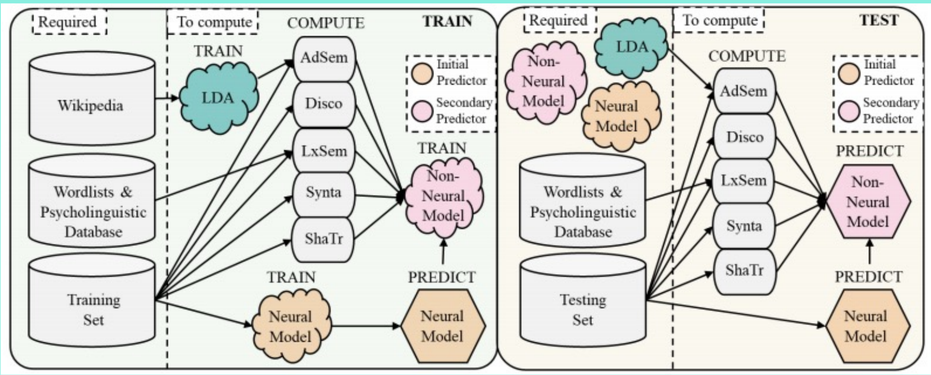
3 - Current Trends



4 - Novel Advanced Semantic Features



8 - Hybrid: Transformer (Initial Predictor) + Traditional ML (2nd Predictor) + Handcrafted Linguistic Features



5 - Testing Transformers

Corpus		BERT	RoBERTa	BART	XLNet
Largest WeeBit	Accuracy	0.893	**0.900**	0.889	0.881
	F1	0.893	**0.900**	0.889	0.880
	Precision	0.896	**0.902**	0.892	0.881
	Recall	0.896	**0.902**	0.892	0.881
	QWK	0.966	**0.970**	0.963	0.966
OneStopE	Accuracy	0.801	0.965	**0.968**	0.804
	F1	0.793	0.965	**0.968**	0.794
	Precision	0.815	0.968	**0.970**	0.810
	Recall	0.814	0.968	**0.970**	0.810
	QWK	0.840	0.942	**0.952**	0.845
Smallest Cambridge	Accuracy	0.573	**0.680**	0.620	0.573
	F1	0.517	**0.658**	0.598	0.554
	Precision	0.528	**0.693**	0.643	0.591
	Recall	0.525	**0.693**	0.643	0.591
	QWK	0.809	**0.881**	0.835	0.832

6 - Testing Traditional ML

Corpus		SVM	RandomF	XGBoost	LogR
WeeBit	Accuracy	**0.679**	0.638	0.638	0.622
	F1	**0.672**	0.626	0.627	0.615
	Precision	**0.696**	0.645	0.656	0.676
	Recall	**0.679**	0.638	0.638	0.622
	QWK	**0.716**	0.703	0.692	0.640
OneStopE	Accuracy	0.737	0.709	0.719	**0.778**
	F1	0.730	0.706	0.701	**0.770**
	Precision	0.751	0.726	0.734	**0.778**
	Recall	0.737	0.709	0.719	**0.778**
	QWK	0.400	0.434	0.363	**0.486**
Cambridge	Accuracy	0.627	0.673	**0.685**	0.680
	F1	0.613	0.663	**0.681**	0.657
	Precision	0.660	0.696	**0.701**	0.694
	Recall	0.627	0.673	**0.674**	0.680
	QWK	0.857	0.880	**0.852**	0.855

7 - Testing Features

Set	Features	LogR	SVM
T1	**AdSem + Disco + Synta + LxSem + ShaTr**	**0.622**	**0.679**
T2	Disco + Synta + LxSem + ShaTr	0.528	0.546
T3	AdSem + Synta + LxSem + ShaTr	0.591	0.582
H1	**AdSem + Disco**	**0.463**	**0.513**
L1	Synta + LxSem	0.499	0.561
L2	**Set L1 - PhrF**	**0.539**	**0.577**
L3	Set L1 - VarF	0.529	0.551
L4	Set L1 - POSF	0.449	0.551
E1	AdSem + PsyF + WorF + TraF	0.489	0.473
E2	**AdSem + PsyF + WorF**	**0.490**	**0.479**
E3	PsyF + WorF	0.464	0.459
P1	EnDF + ShaF + TrSF + POSF + WorF + PsyF + TraF	0.608	0.633
P2	Set P1 + TraF	0.629	0.638
P3	**Set P2 + VarF**	**0.647**	**0.674**

9 - Hybrid Model Results

Corpus		Model																	
		Baselines, Previous Studies				BERT			RoBERTa			BART			XLNet		
		Xia-16	Fili-19	Mar-21		hybrid	Δ	Δ	hybrid	Δ	Δ	hybrid	Δ	Δ	hybrid	Δ	Δ
		SVM	LSTM	BERT	HAN	GB-T1	BERT	GB	RF-T1	RBRT	RF	RF-T1	BART	RF	RF-P3	XLNet	RF
WeeBit	Accuracy	0.803	0.813	**0.857**	0.752	0.895	0.002	0.257	0.902	0.002	0.264	**0.905**	0.016	0.267	0.892	0.011	0.254
	F1	-	-	**0.866**	0.753	0.895	0.002	0.268	0.902	0.002	0.276	**0.905**	0.016	0.279	0.892	0.012	0.266
	Precision	-	-	**0.857**	0.752	0.897	0.001	0.241	0.903	0.001	0.258	**0.905**	0.013	0.260	0.893	0.012	0.248
	Recall	-	-	**0.858**	0.752	0.897	0.001	0.259	0.903	0.001	0.265	**0.904**	0.012	0.266	0.892	0.011	0.254
	QWK	-	-	**0.953**	0.886	0.969	0.001	0.277	0.971	0.001	0.268	**0.968**	0.005	0.265	0.966	0.000	0.263
OneStopE	Accuracy	-	-	0.674	**0.787**	0.982	0.181	0.263	**0.990**	0.025	0.281	0.971	0.003	0.262	0.848	0.044	0.139
	F1	-	-	0.740	**0.798**	0.982	0.189	0.281	**0.995**	0.030	0.289	0.971	0.003	0.265	0.848	0.050	0.142
	Precision	-	-	0.674	**0.787**	0.983	0.168	0.249	**0.995**	0.027	0.269	0.972	0.002	0.246	0.852	0.042	0.126
	Recall	-	-	0.677	**0.789**	0.982	0.168	0.263	**0.996**	0.028	0.287	0.971	0.001	0.262	0.848	0.038	0.139
	QWK	-	-	0.708	**0.825**	0.973	0.133	0.610	**0.996**	0.054	0.562	0.952	0.000	0.518	0.855	0.010	0.369
Cambridge	Accuracy	0.786**	-	-	-	0.687	0.114	0.002	**0.763**	0.083	0.090	0.727	0.107	0.054	0.687	0.114	0.014
	F1	-	-	-	-	0.682	0.165	0.001	**0.752**	0.094	0.089	0.727	0.129	0.064	0.676	0.122	0.013
	Precision	-	-	-	-	0.732	0.204	0.031	**0.792**	0.099	0.096	0.760	0.117	0.064	0.710	0.119	0.014
	Recall	-	-	-	-	0.687	0.162	0.013	**0.753**	0.060	0.080	0.727	0.084	0.054	0.687	0.096	0.014
	QWK	-	-	-	-	0.873	0.064	0.021	**0.919**	0.038	0.039	0.889	0.054	0.009	0.888	0.056	0.008

1. **RoBERTa-RandomForest-T1** hybrid shows consistently high performance on all metrics. The model achieved the near-perfect accuracy of 99% on OneStopEnglish. Considering that the previously reported accuracies have never exceeded 80%, the new SOTA is an exceptional improvement.

2. For a medium-sized dataset, both neural and non-neural models benefit from the hybrid architecture. **BERT-GradientBoost-T1** hybrid showed 98.2% accuracy on OneStopEnglish. This is a 18.1% increase from BERT-only performance and a 26.3% increase from GradientBoost-only performance.

3. Overall, hybrid architecture did not benefit neural model (transformer) for WeeBit dataset. WeeBit is the largest available dataset (625 items, 5 classes) in readability assessment.

4. Overall, hybrid architecture largely benefit (~10%) neural model (transformer) for Cambridge dataset. Cambridge is the smallest dataset we tested (60 items, 5 classes)

5. **BERT-GradientBoost-T1** achieves 85.8% accuracy on WeeBit using only 750 passages, 30% of the original train data. For comparison, 85.7% was the past SOTA (using BERT only).